

How to identify normal forms

How to Find Minimal Cover

Steps:

1. Re-structure all Fds so that they have single attribute on RHS.
 - Hint: Use Decomposition Rule.
2. Pick those Fds which have more than one attribute on L.H.S
 - Take closure of every single attribute which is on LHS of Fd.
 - If the closure yields all attributes that were on RHS then restructure the fd to include only one attribute on LHS.
3. Check all the FDs if there is a situation of transitivity.
 - For example if we have Fds $a \rightarrow b$, $b \rightarrow c$ & $a \rightarrow c$. We can remove $a \rightarrow c$ from the set of fds as it can be inferred any time using the other two fds.

Example 1:

- $AB \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow EF$
- $A \rightarrow CF$

Step 1

- $AB \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

Step 2

- $AB \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

Step 2a

- Taking $AB \rightarrow D$
- Taking Closure
- $A^+ \rightarrow D$
- $B^+ \rightarrow B$
- So we can minimize this FD to
- $A \rightarrow D$

Example 1:

Step 2

- $AB \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

Step 2a

- Taking $AB \rightarrow D$
- Taking Closure
- $A^+ \rightarrow D$
- $B^+ \rightarrow B$
- So we can minimize this FD to
- $A \rightarrow D$

Step 2b

Taking $AE \rightarrow B$

Taking Closure

$A^+ \rightarrow$ doesnot have B

$E^+ \rightarrow$ doesnot have B

So this FD will remain as is.

After Completion of step 2 we have following FDs

- $A \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

Example 1:

After Completion of step 2 we have following FDs

- $A \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

• Next Step is to remove Duplicates & identities.

- $A \rightarrow D$
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

After duplicate removal:

- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

Example 1:

Step 3

$B \rightarrow C$
 $AE \rightarrow B$
 $A \rightarrow D$
 $D \rightarrow E$
 $D \rightarrow F$
 $A \rightarrow C$
 $A \rightarrow F$

- Now we need to find if there is any FD which can be transitively inferred. We will remove them.
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$
- $A \rightarrow F$

In this case $A \rightarrow F$ should be removed.

- The Minimal Cover would be:
- $B \rightarrow C$
- $AE \rightarrow B$
- $A \rightarrow D$
- $D \rightarrow E$
- $D \rightarrow F$
- $A \rightarrow C$

In this case $A \rightarrow F$ should be removed.

Example 2:

- Example 2: Let the given set of FDs be $G: \{A \rightarrow BCDE, CD \rightarrow E\}$.

■ Here, the given FDs are NOT in the canonical form. So we first convert them into: $E: \{A \rightarrow B, A \rightarrow C, A \rightarrow D, A \rightarrow E, CD \rightarrow E\}$.

■ In step 2 of the algorithm, for $CD \rightarrow E$, neither C nor D is extraneous on the left-hand side, since we cannot show that $C \rightarrow E$ or $D \rightarrow E$ from the given FDs. Hence we cannot replace it with either.

■ In step 3, we want to see if any FD is redundant. Since $A \rightarrow CD$ and $CD \rightarrow E$, by transitive rule (IR3), we get $A \rightarrow E$. Thus, $A \rightarrow E$ is redundant in G .

Example 2:

• ■ So we are left with the set F, equivalent to the original set G as: $\{A \rightarrow B, A \rightarrow C, A \rightarrow D, CD \rightarrow E\}$. F is the minimum cover. As we pointed out in footnote 6, we can combine the first three FDs using the union rule (IR5) and express the minimum cover as:

Minimum cover of G, F: $\{A \rightarrow BCD, CD \rightarrow E\}$.

Example 2:

• ■ So we are left with the set F, equivalent to the original set G as: $\{A \rightarrow B, A \rightarrow C, A \rightarrow D, CD \rightarrow E\}$. F is the minimum cover. As we pointed out in footnote 6, we can combine the first three FDs using the union rule (IR5) and express the minimum cover as:

Minimum cover of G, F: $\{A \rightarrow BCD, CD \rightarrow E\}$.

Example 3:

Let the given set of FDs be $E: \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$. We have to find the minimal cover of E .

- All above dependencies are in canonical form (that is, they have only one attribute on the right-hand side), so we have completed step 1 of Algorithm 15.2 and can proceed to step 2. In step 2 we need to determine if $AB \rightarrow D$ has any redundant (extraneous) attribute on the left-hand side; that is, can it be replaced by $B \rightarrow D$ or $A \rightarrow D$?
- Since $B \rightarrow A$, by augmenting with B on both sides (IR2), we have $BB \rightarrow AB$, or $B \rightarrow AB$ (i). However, $AB \rightarrow D$ as given (ii).
- Hence by the transitive rule (IR3), we get from (i) and (ii), $B \rightarrow D$. Thus $AB \rightarrow D$ may be replaced by $B \rightarrow D$.

Example 3:

■ We now have a set equivalent to original E , say E' : $\{B \rightarrow A, D \rightarrow A, B \rightarrow D\}$. No further reduction is possible in step 2 since all FDs have a single attribute on the left-hand side.

■ In step 3 we look for a redundant FD in E' . By using the transitive rule on $B \rightarrow D$ and $D \rightarrow A$, we derive $B \rightarrow A$. Hence $B \rightarrow A$ is redundant in E' and can be eliminated.

■ Therefore, the minimal cover of E is F : $\{B \rightarrow D, D \rightarrow A\}$.